Kun-Ying Lee

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Robot Learning | Humanoid Locomotion | Motion Generation

Education

Tsinghua University Sept. 2025–Jun. 2029 (expected)
Undergraduate, Yao Class GPA: 3.8/4.0
Computer Science (Artificial Intelligence Track)
Institute for Interdisciplinary Information Sciences (IIIS)

Selected Publications

OMG: Omni-Modal Motion Generation for Generalist Humanoid Control
Siqiao Huang*, **Kun-Ying Lee***, Dongming Qiao*, Guanqi He*, Zhenyu Wang, Yitang Li, Shaoting Zhu, and Hang Zhao.
arXiv:2606.10340, 2026. Paper | Project | Code
RSS 2026 workshops: ExWBC (**Oral**); RoboData (**Spotlight**).

Hiking in the Wild: A Scalable Perceptive Parkour Framework for Humanoids
Shaoting Zhu*, Ziwen Zhuang*, Mengjie Zhao, **Kun-Ying Lee**, and Hang Zhao.
Conference on Robot Learning (CoRL), 2026 (accepted). Paper | Project | Code

The Indefinite Integral of the *n*th Power of Inverse Trigonometric Functions
Kun-Ying Lee.
Mathmedia, Academia Sinica, 49(2):91–101, 2025. Article

* Equal contribution.

Research Experience

MARS Lab, Tsinghua University Nov. 2025–present
Research Assistant | Advisor: Prof. Hang Zhao

OMG-Posttrain Project Lead; ongoing

- Lead research on reinforcement learning post-training of a motion generator to enable a frozen motion tracker to execute highly dynamic skills beyond its baseline capabilities.
- Built an end-to-end residual PPO pipeline around a frozen motion generator and tracker, integrating state-conditioned reference corrections, vectorized closed-loop rollouts, and independent execution-based evaluation.
- Investigating diffusion-based reinforcement learning to directly fine-tune the full motion-generation backbone using closed-loop execution feedback.

OMG: Omni-Modal Motion Generation Co-first author

- Built a video-to-robot data pipeline combining GVHMR motion reconstruction, humanoid retargeting, and VLM-based motion captioning to convert monocular dance videos into annotated robot-motion training data.
- Developed and pretrained a multimodal diffusion-transformer motion generator conditioned on language, audio, and human motion; conducted ablation studies of multimodal conditioning and attention mechanisms.
- Built an asynchronous closed-loop deployment system on Unitree G1, integrating ONNX/TensorRT inference, replanning from measured robot-state history, and temporal alignment between motion generation and control.

Hiking in the Wild: Perceptive Humanoid Parkour

- Validated sim-to-sim transfer of a depth-conditioned parkour policy to support deployment on Unitree G1.
- Resolved a sim-to-real perception gap caused by reflection-induced depth artifacts through invalid-depth masking, spatial inpainting, and Gaussian filtering, aligning real sensor inputs with training observations.
- Supported real-robot deployment and field testing; the system traversed stairs, slopes, and gaps, with running speeds up to 2.5 m/s and gap widths up to 50 cm.

Engineering Experience

TH-MOS Robot Soccer Team, Tsinghua University Sept. 2025–present

Gait Control Group Lead | *Decision-Making Algorithms Lead*

K1Lab: Learning and Control

- Developed command-conditioned adversarial motion priors (AMP) for omnidirectional locomotion, achieving stable hardware tracking over forward/backward velocities of $[-1.5, +2.0]$ m/s, lateral velocities of $[-1.5, +1.5]$ m/s, and yaw rates of $[-1.7, +1.7]$ rad/s on Booster K1.
- Developed target-conditioned shooting policies targeting specified 3D coordinates within the goal mouth from positions across a 14×9 m field, achieving targeting errors within 10 cm in 90% of physical trials (27/30); currently exploring onboard-perception-guided curved shots for obstacle avoidance.
- Trained an AMP-based fall-recovery policy in MJLab that recovers from tested fallen poses to stable standing within 0.5 s, succeeding in all 30 real-robot trials.

Robot Sports: Simulation and Evaluation

- Developed a Unity–MuJoCo humanoid soccer platform supporting autonomous 3v3 and 5v5 matches, integrating learned locomotion, kicking, and recovery controllers with the HSL GameController.
- Designed modular decision-to-skill interfaces and per-robot perception pipelines combining YOLO-based object detection with particle-filter-based localization.
- Investigating self-play multi-agent reinforcement learning for team decision-making, starting from MAPPO.

MOS-9 Open-Source Humanoid Robot Project

- Contributed to robot deployment, AMP-based locomotion training in Isaac Lab, motion-data retargeting, and MuJoCo sim-to-sim validation for mos-9.
- Contributed to lightweighting with carbon fiber, reducing robot mass from 16 kg to 11 kg.
- The project received the **Special Prize** at the National Additive Manufacturing Competition.

Open-Source Projects

SuperMimic on mjlab: Model-Based Humanoid Motion Tracking Paper | Code

- Reproduced SuperMimic's differentiable-world-model approach to humanoid motion tracking on native MuJoCo-Warp through mjlab, achieving 100% full-clip completion across 40 LAFAN1 motions within 5 hours of training on a single RTX 5090.
- Demonstrated few-shot adaptation of the pretrained tracking policy to highly dynamic MotionCode motions through fine-tuning on a small motion set; implemented mixed-frame-rate data processing and state-preserving checkpoint migration.

ScaleBFM-K1: Large-Scale Humanoid Motion Tracking Code

- Adapted ScaleBFM-based motion tracking to the 22-DoF Booster K1, achieving over 95% full-clip completion on the BONES-SEED test set after 5 hours of training on one node with 8 NVIDIA A10 GPUs.
- Built and validated a pipeline for 142,220 BONES-SEED and 77 LAFAN1 motion clips, including morphology-consistent conversion and differentiable forward kinematics verified against MuJoCo.

Selected Honors & Awards

Ministry of Education Scholarship for Taiwan Students (**First Prize, Top 2 Schoolwide**) Oct. 2026

5th Place, RoboCup 2026 Incheon (TH-MOS team) Jul. 2026

2nd Place, RoboCup Asia-Pacific Tianjin Invitational Tournament (550W team) May. 2026

3rd Place, RoboCup China Open (TH-MOS team) May. 2026

Leading Talent (Lingjun) Award, Tsinghua University Oct. 2025